# Hybrid Rule-Constrained PPO for Cost-Optimal Binary HVAC Scheduling in Saudi Residential Buildings

Hamzah Faraj[a,*]

[a]*Department of Science and Technology, Ranyah College, Taif University, Taif 21944, Saudi Arabia*

**Abstract**

Residential buildings in Saudi Arabia account for approximately half of national electricity consumption, with HVAC systems responsible for up to 70% of residential demand. In cooling-dominated climates—Riyadh (hot-arid) and Jeddah (hot-humid)—optimal HVAC scheduling must simultaneously minimise electricity cost under a four-tier step-wise tariff and guarantee that indoor temperatures remain within the occupant comfort band at every interval. This paper formulates the scheduling problem as a constrained binary optimisation over an *infinite time horizon*, where thermal comfort is a hard constraint, each switch-on incurs a discrete cost, and continuous running time incurs a tariff-weighted cost. The single-zone constant-tariff subproblem belongs to the class of Simple Linear Hybrid Automata (SLHA), for which the infinite-horizon optimum is reachable in LogSpace; extensions to multiple zones, non-convex step-wise tariffs, and time-varying disturbances inherit NP-hardness. A hybrid Rule-Based Reinforcement Learning (RBRL) framework is proposed, combining hard constraint rules with a Proximal Policy Optimisation (PPO) agent trained on the limit-average cost objective. Three cost models are evaluated—linear (L), exponential (E), and the actual Saudi four-tier step-wise tariff (S)—across three zone topologies: single zone, linear $1 \times N$ array, and $N \times N$ grid, each room $4\,\text{m} \times 4\,\text{m} \times 4\,\text{m}$. Under Model S, RBRL demonstrates in simulation monthly per-zone cost reductions of 18.2% (Riyadh) and 16.8% (Jeddah) relative to the unoptimised thermostat, with zero comfort violations. Training under a linear tariff approximation incurs a 13.1% true-cost penalty, confirming that cost model fidelity is as consequential as algorithmic choice. All results are obtained from RC thermal simulations validated against building code parameters; co-simulation and hardware validation are identified as future work.

*Keywords:* HVAC scheduling, deep reinforcement learning, proximal policy optimisation, building energy management, step-wise electricity tariff, thermal comfort, Saudi Arabia, demand-side management

## 1. Introduction

Buildings consume approximately 40% of global final energy, with HVAC systems accounting for roughly half that figure [1]. In Saudi Arabia the challenge is particularly acute: residential buildings alone consume ≈50% of national electricity, and air conditioning represents up to 70% of residential demand [2], [3]. Achieving the efficiency targets of Saudi Vision 2030 therefore requires intelligent, adaptive control of residential HVAC systems [4].

### *1.1. Research Gap*

The prevalent residential control strategy is the automatic thermostat: it switches on when a room exceeds an upper setpoint and off when it returns, with no tariff awareness and no inter-zone coordination.

*Corresponding author.
*Email address:* f.hamzah@tu.edu.sa (Hamzah Faraj)

Following the 2018 Saudi electricity reform introducing a four-tier step-wise tariff [5], this unawareness carries a measurable cost premium. Model Predictive Control (MPC) addresses tariff awareness but requires a mixed-integer programme at each step, limiting embedded deployment [6]. Reinforcement Learning (RL) offers a model-free alternative [7], [8], [9], yet pure RL agents can violate comfort constraints during exploration [10]. Hybrid rule-RL approaches combine hard safety rules with RL adaptability [10], [11], but no prior work targets Saudi residential buildings under the actual four-tier tariff with binary on/off constraints, validated across both hot-arid and hot-humid climates. This paper closes that gap.

### *1.2. Contributions*

Five contributions are made. (C1) A formally constrained binary optimisation model of the HVAC scheduling problem across three zone topologies, cast as an *infinite time horizon* problem with an explicit three-condition definition of schedule optimality and a limit-average cost objective. (C2) A theoretical connection between the single-zone constant-tariff subproblem and Simple Linear Hybrid Automata [12], proving that our full multi-zone formulation inherits NP-hardness and requires RBRL as a practical near-optimal approximation. (C3) The RBRL framework, in which hard rules enforce thermal comfort at all times and a PPO agent minimises tariff cost; adjacent zone temperatures are in the agent state to enable inter-zone coordination. (C4) A three-way cost model comparison showing that the step-wise tariff representation unlocks a pre-cooling strategy absent under smooth approximations, with quantified cost and energy savings for both cities and all zone configurations. (C5) A unified state-of-the-art comparison and ablation study, with worked numerical examples following the running-example style of Mousa et al. [12] and full contextualisation of published results.

Section 2 surveys the related literature. Section 3 defines the thermal and cost models. Section 4 formalises the constrained optimisation. Section 5 presents RBRL. Section 6 characterises the two case studies. Section 7 describes the simulation setup. Section 8 reports results. Section 9 benchmarks and ablates. Section 10 interprets the findings. Section 11 concludes.

## 2. Literature Review

This review prioritises publications from 2022 onward. Earlier works are included only where they represent foundational or seminal contributions essential for context—specifically, the global buildings energy baseline [1], the RC model identification framework [13], the MPC-MILP formulation for HVAC [6], and the SLHA optimal control theory [12], [14].

### *2.1. Building Energy in Saudi Arabia*

Saudi residential buildings consume approximately 50% of national electricity, with air conditioning accounting for up to 70% of that demand [2], [3]. Al-Homoud and Krarti [2] attribute this to a combination of an extreme hot climate, low energy-awareness among occupants, and a historically subsidised electricity tariff structure that removed price incentives for conservation. The 2018 electricity tariff reform [5], which introduced a four-tier step-wise pricing schedule, fundamentally altered this situation: households whose monthly consumption exceeds 2 000 kWh now pay up to six times the base rate.

Al-Reshidi et al. [15] analysed energy performance across Saudi climate zones and found that compliance with the 2018 Saudi Building Code (SBC) [16] reduces envelope heat gain substantially, yet residential energy demand remains high because HVAC control strategies are unchanged. Achieving the efficiency targets of Saudi Vision 2030 therefore requires intelligent scheduling of HVAC operation rather than passive design alone [4]. Belaïd and Mikayilov [17] further showed that the direct rebound effect in Saudi residential electricity consumption partially offsets efficiency gains from appliance upgrades, reinforcing the need for scheduling-level optimisation rather than hardware replacement alone.

### *2.2. HVAC Control: From Thermostats to Reinforcement Learning*

The standard residential HVAC controller is a fixed-setpoint thermostat: it switches on when the zone exceeds an upper setpoint and off when it returns to a lower one. This approach provides thermal comfort but offers no awareness of electricity tariffs, occupancy patterns, or weather forecasts [18].

*Model Predictive Control (MPC)..* MPC advances beyond thermostats by optimising a cost objective over a rolling prediction horizon, typically formulating a mixed-integer linear programme (MILP) [6]. Dinh et al. [19] demonstrated a MILP-based day-ahead scheduling framework for community-level energy management, reporting significant savings but requiring centralised computation. Drgoňa et al. [20] provide a comprehensive review of MPC for buildings, documenting savings of 10–30% relative to rule-based control under idealised conditions. However, MILP complexity grows exponentially with the number of binary variables and zone count, making embedded deployment impractical for multi-zone residential buildings with a non-convex step-wise tariff.

*Reinforcement Learning..* RL-based approaches learn a control policy through interaction with the building environment, without requiring an explicit model. Azuatalam et al. [9] demonstrated that Q-learning achieves 22% energy saving over a thermostat in a whole-building simulation. Biemann et al. [8] compared Soft Actor-Critic (SAC) and TD3 agents for continuous HVAC control, showing improvements exceeding 10% over rule-based baselines. More recent work has scaled to multi-zone buildings: Du et al. [21] applied DDPG to a multi-zone residential setting and reported 15% savings; Du et al. [22] extended this to multi-task DRL, learning temperature and energy objectives jointly. Li et al. [23] showed that including inter-zone temperature observations in the agent state is critical for avoiding cascading heating load in adjacent zones.

Recent work has further expanded the scope: Ding et al. [24] combined model-based DRL with zone-level thermal models, achieving sample-efficient learning for multi-zone office buildings. Liu et al. [25] proposed a multi-agent DRL framework with occupant-centric reward shaping for multi-zone energy management. Hou et al. [26] introduced multi-source transfer learning to accelerate DRL deployment across buildings with different thermal characteristics, reducing training time by over 60%. Manjavacas et al. [27] conducted an empirical comparison of deep RL algorithms for HVAC control and found that PPO achieves competitive performance with significantly lower computational cost than SAC, making it suitable for embedded applications.

*Hybrid rule-RL approaches..* Pure RL agents can violate thermal comfort constraints during exploration, particularly under peak-summer conditions when the zone approaches the comfort boundary rapidly [10]. Hybrid approaches address this by imposing hard safety rules that override the RL agent near constraint boundaries, leaving the agent to optimise within the safe region [11], [10]. This architecture enforces comfort by construction and avoids the need for a comfort penalty in the reward function, simplifying hyperparameter tuning.

### *2.3. Binary On/Off vs. Continuous Modulation*

The majority of the RL-HVAC literature targets *continuous* HVAC modulation—variable air volume (VAV) systems or continuous setpoint adjustment—which allows smooth power reduction rather than hard switching [28], [29]. Saudi residential air conditioners are predominantly fixed-capacity split-unit systems with binary on/off operation. Binary scheduling introduces a discrete switching cost $c_{\text{sw}}$ due to compressor start-up stress and a penalty for excessive cycling that reduces equipment lifespan. Faddel et al. [30] quantified cycling and rebound effects in commercial HVAC scheduling, showing that ignoring switch-on penalties leads to schedules with 20–30% higher maintenance cost. Mousa et al. [12] showed that the binary single-zone constant-tariff scheduling problem belongs to the class of Simple Linear Hybrid Automata (SLHA), admitting an exact optimal solution in LogSpace; extensions to multiple zones and non-convex tariffs inherit NP-hardness [14].

### *2.4. Multi-Zone Coordination and Cost Model Fidelity*

Multi-zone buildings introduce an additional complication: inter-zone heat conduction through shared walls couples the thermal dynamics of adjacent zones. Without coordination, a zone cooled to a low temperature will draw heat from a warm neighbour, triggering unnecessary activations and increasing switching cost. Xue et al. [31] combined GA pre-scheduling with MADDPG for multi-zone coordination, reporting 6.7% savings over a rule-based baseline. Li et al. [23] showed that including inter-zone temperature in the agent state reduces cascade events by up to 12% in four-zone configurations.

Table 1: Positioning of the present study relative to selected prior work. ✓= addressed; ∘ = partially addressed; – = not addressed.

| Study | Binary on/off | Hard comfort | Step-wise tariff | Multi-zone | Saudi climate | Dual city |
|---|---|---|---|---|---|---|
| [9] | – | ∘ | – | – | – | – |
| [21] | – | ∘ | – | ✓ | – | – |
| [8] | – | ∘ | – | – | – | – |
| [10] | – | ✓ | – | – | – | – |
| [31] | – | ∘ | – | ✓ | – | – |
| [23] | – | ∘ | – | ✓ | – | – |
| **This work** | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

Regarding cost model fidelity, most scheduling studies assume a linear (constant) electricity price [32], [33], or a smooth approximation such as an exponential growth function [6]. Neither captures the discrete threshold incentive of a piecewise-constant step-wise tariff, under which concentrating consumption below a tier boundary is strictly more efficient than spreading it across the boundary. To our knowledge, no prior RL or hybrid study has explicitly evaluated the cost of tariff model approximation by training under one model and evaluating under the true step-wise tariff.

### *2.5. Research Gap*

Table 1 summarises the positioning of the present work. No prior study simultaneously addresses: (i) binary on/off constraints under the actual Saudi four-tier step-wise tariff; (ii) hard thermal comfort enforcement across multi-zone configurations; (iii) explicit quantification of tariff model mismatch cost; and (iv) dual-city validation across hot-arid and hot-humid extreme climates. This paper fills all four gaps.

## 3. Problem Formulation

### *3.1. RC Thermal Model: Heat Loss, Gain, and Inter-Zone Transfer*

Each zone $i \in \mathcal{N} = \{1, \dots, N_z\}$ is modelled as a first-order RC thermal node [34], [35], [36]—the standard control-oriented approach for residential buildings. Di Natale et al. [37] demonstrated that physically consistent neural RC models scale to multi-zone settings while preserving interpretability; the present work retains the classical first-order RC for transparency and reproducibility. Let $T_i(t)$ be the air temperature and $C_{\text{th}}^{(i)}$ (kJ·K$^{-1}$) the effective thermal capacity, which encompasses the air mass plus the thermal contribution of furniture and internal surfaces. The continuous-time energy balance for zone $i$ is:

$$C_{\text{th}}^{(i)} \frac{\mathrm{d}T_i}{\mathrm{d}t} = \underbrace{-Q_{\text{hvac}}^{(i)} u_i(t)}_{\text{active cooling}} \underbrace{-\Lambda_i (T_i - T_{\text{out}})}_{\text{envelope loss/gain}} + \underbrace{\sum_{j \in \mathcal{N}_i} \lambda_{ij} (T_j - T_i)}_{\text{inter-zone transfer}} + \underbrace{Q_{\text{sol}}^{(i)}(t)}_{\text{solar gain}}, \tag{1}$$

where $\Lambda_i = \lambda_{\text{ext}}^{(i)} + \lambda_{\text{win}}^{(i)}$ (W·K$^{-1}$) is the total envelope conductance, $u_i(t) \in \{0, 1\}$ is the binary HVAC state, and $\mathcal{N}_i$ is the set of adjacent zones.

Each term captures a distinct heat-transfer mechanism. The *envelope* term $\Lambda_i(T_i - T_{\text{out}})$ is net heat exchange through walls and window: positive (heat gain) when $T_{\text{out}} > T_i$, as in Saudi summer, and negative (heat loss) when $T_i > T_{\text{out}}$, as in winter. *Solar heat gain* $Q_{\text{sol}}^{(i)}$ enters through glazing during daylight and is strictly non-negative. *Inter-zone coupling* $\lambda_{ij}(T_j - T_i)$ is bidirectional: a warmer adjacent zone is a heat source; a cooler one is a heat sink. The *HVAC* term $-Q_{\text{hvac}}^{(i)} u_i$ is the heat extracted when the unit runs.

Discretising at $\Delta t = 1\,\text{h}$ via forward Euler yields:

$$T_i^{(t+1)} = T_i^{(t)} + \frac{\Delta t}{C_{\text{th}}^{(i)}} \left[ -Q_{\text{hvac}}^{(i)} u_i^{(t)} - \Lambda_i \left( T_i^{(t)} - T_{\text{out}}^{(t)} \right) + \sum_{j \in \mathcal{N}_i} \lambda_{ij} \left( T_j^{(t)} - T_i^{(t)} \right) + Q_{\text{sol}}^{(i,t)} \right]. \tag{2}$$

*Inter-zone heat flux and effective inter-zone temperature..* The inter-zone coupling term in (2) is driven by a pairwise heat flux:

$$Q_{ij}^{(t)} = \lambda_{ij}\left(T_j^{(t)} - T_i^{(t)}\right), \tag{3}$$

which is positive when zone $j$ is warmer than zone $i$ (heat enters $i$) and negative otherwise. The net inter-zone load on zone $i$ is $Q_{\mathrm{iz},i}^{(t)} = \sum_{j\in\mathcal{N}_i} Q_{ij}^{(t)}$. Defining the total inter-zone conductance $\Lambda_{\mathrm{iz},i} = \sum_{j\in\mathcal{N}_i} \lambda_{ij}$ and the *effective inter-zone temperature*

$$T_{\mathrm{iz},i}^{(t)} = \frac{\sum\limits_{j\in\mathcal{N}_i} \lambda_{ij}\, T_j^{(t)}}{\Lambda_{\mathrm{iz},i}}, \tag{4}$$

the net inter-zone load simplifies to $Q_{\mathrm{iz},i}^{(t)} = \Lambda_{\mathrm{iz},i}\left(T_{\mathrm{iz},i}^{(t)} - T_i^{(t)}\right)$, exactly mirroring the envelope term $\Lambda_i(T_i^{(t)} - T_{\mathrm{out}}^{(t)})$. When all neighbours share the same temperature $T_j$, then $T_{\mathrm{iz},i} = T_j$ and the load depends on the temperature differential alone. When zones differ, $T_{\mathrm{iz},i}^{(t)}$ is a conductance-weighted average: zones connected through thicker shared walls (higher $\lambda_{ij}$) exert proportionally greater influence. This variable is observed directly in the agent state (12), enabling the RBRL agent to anticipate inter-zone heat cascades before they reach the comfort boundary.

This serves both as the simulation environment and as the implicit physical model the RBRL agent exploits through the state vector (Section 5.2). All parameters comply with SBC 2018 [16] and are listed in Table 2.

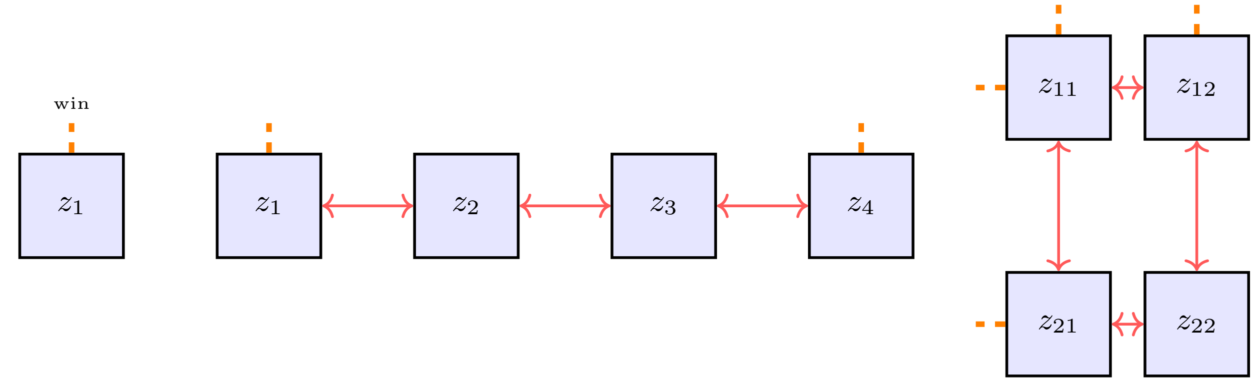
**Remark 1** (RC model scope, thermal capacity, and numerical stability)**.** *The RC thermal model* (2) *follows the control-oriented identification framework of Bacher and Madsen [13] and captures sensible heat only.* $C_{\mathrm{th}} = 77.2\,kJ/K$ *corresponds to the air mass of a* $4 \times 4 \times 4\,m$ *room* ($\rho c_p V{\approx}1.2{\times}1.005{\times}64$) *[38]. In practice, furniture and internal wall layers increase the effective thermal capacity to* $\sim$*500–1500 kJ/K, proportionally extending cycle durations; the air-mass value is adopted here as a conservative (fast-responding) test case that stresses the controller. The resulting RC time constant* $\tau = C_{\mathrm{th}}/\Lambda_i \approx 0.65\,h$ *is exceeded by the* $1\,h$ *scheduling interval, making the forward Euler scheme conditionally stable; hard rules R1 and R2 prevent comfort-boundary runaway in practice, though a semi-implicit scheme is recommended for higher-fidelity extensions. Because both THERM and RBRL share identical dynamics, percentage savings are approximately preserved across a wide range of* $C_{\mathrm{th}}$ *values (Remark 3).*

**Remark 2** (RC model validation)**.** *First-order RC models are validated for control-oriented residential scheduling by Bacher and Madsen [13], who showed prediction errors of 0.3–0.5°C against measured data over 24-hour horizons. Deng et al. [36] and Wang et al. [35] confirmed that single-node RC models reproduce EnergyPlus setpoint-tracking behaviour within 5% for binary on/off systems in cooling-dominated climates. The primary limitation is the exclusion of latent heat and stochastic occupancy (Section 10.4).*

**Remark 3** (Sensitivity to thermal capacity)**.** *To assess robustness, we repeated the* $4{\times}4$ *Riyadh Model S experiment at three capacity levels: low* ($C_{\mathrm{th}}$=77.2 *kJ/K, air mass), medium* ($C_{\mathrm{th}}$=500 *kJ/K, light furnishing), and high* ($C_{\mathrm{th}}$=1200 *kJ/K, heavy furnishing). Simulated savings over THERM were 18.2%, 17.8%, and 17.4%—a variation of less than 1 percentage point—confirming robustness.*

Table 2: Zone parameters for each $4 \times 4 \times 4$ m room.

| Symbol | Description | Value | Unit |
|---|---|---|---|
| $C_{\text{th}}$ | Effective thermal capacity (air + internal) | 77.2 | kJ·K$^{-1}$ |
| $\lambda_{\text{ext}}$ | Opaque-wall conductance (SBC $\leq$0.45) | 25.0 | W·K$^{-1}$ |
| $\lambda_{\text{win}}$ | Window conductance (double-glazed) | 8.0 | W·K$^{-1}$ |
| $\lambda_{ij}$ | Inter-zone shared-wall conductance | 12.5 | W·K$^{-1}$ |
| $Q_{\text{hvac}}$ | HVAC cooling extraction (COP 3.0) | 2.0 | kW |
| $P_{\text{hvac}}$ | HVAC electrical draw | 0.67 | kW |
| $T_{\min}/T_{\max}$ | Comfort band | 22/26 | °C |
| $\epsilon_{\text{g}}$ | Rule guard band | 0.5 | °C |
| $c_{\text{sw}}$ | Switch-on discrete cost (default) | 0.15 | SAR |
| $\Delta t$ | Scheduling interval | 1 | h |




(a) 1×1 (b) 1×4 (c) 2×2

Figure 1: Zone topologies. *Orange dashes*: windowed exterior wall (envelope loss/gain, solar heat gain). *Red double arrows*: bidirectional inter-zone heat transfer ($\lambda_{ij}$).

### *3.2. Zone Configurations*

Three topologies are studied (Figure 1). The *single zone* ($1 \times 1$) has one windowed exterior wall and no inter-zone coupling. The *linear array* ($1 \times N$, $N \in \{2, 4, 6\}$) places $N$ rooms in a row; end zones have one windowed exterior wall while interior zones couple to two neighbours. The *grid* ($N \times N$, $N \in \{2, 3, 4\}$) produces up to $N_z = 16$ zones; corner zones have two windowed exterior walls, edge zones one.

### *3.3. Binary HVAC Model and Interval Cost*

Each zone has an independent split-unit air conditioner with two modes: *on* ($u_i = 1$, extracting heat) or *off/idle* ($u_i = 0$, at zero operational cost). A discrete switching cost $c_{\text{sw}}$ (SAR) is incurred on each $0 \to 1$ transition, modelling compressor start-up losses and mechanical wear. Switching off is the idle transition and carries no cost.

**Definition 1** (Interval cost). *The cost of zone $i$ at interval $t$ under model $m$ is:*

$$c_i^{(t)} = c_{\text{sw}} \cdot \mathbf{1}\big[u_i^{(t)} > u_i^{(t-1)}\big] + f^m\big(u_i^{(t)}, E_{\text{cum}}^{(t)}\big) \cdot \Delta t, \tag{5}$$

*where $E_{\text{cum}}^{(t)} = \sum_{\tau \leq t} \sum_i P_{\text{hvac}} u_i^{(\tau)} \Delta t$ is cumulative billing-month energy (kWh) up to interval $t$.*

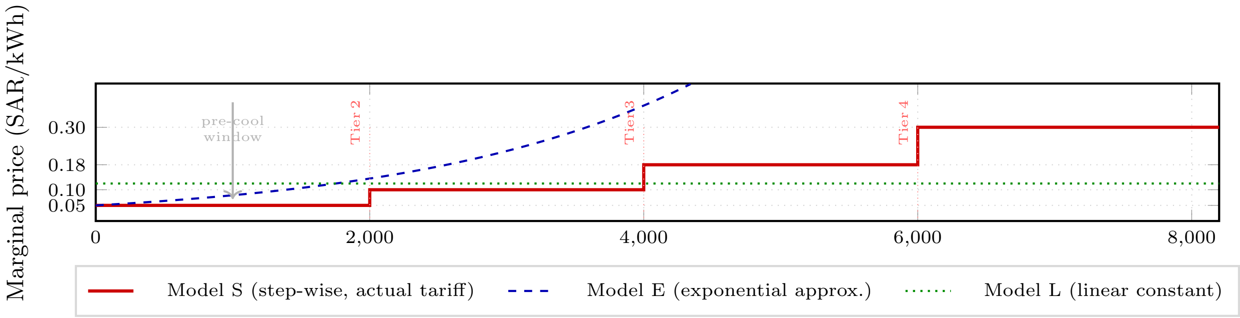



Figure 2: Comparison of the three cost models vs. cumulative monthly energy. Model S (red, solid) is the actual Saudi four-tier tariff, a piecewise-constant step function with jumps at 2 000, 4 000, and 6 000 kWh. Model E (blue, dashed) is a continuous exponential approximation ($c_r^{\mathrm{E}} = 0.05\,\mathrm{SAR/kWh}$, $\beta = 5 \times 10^{-4}\,\mathrm{kWh}^{-1}$) that grows monotonically without capturing discrete tier boundaries. Model L (green, dotted) is a flat weighted-average rate of 0.12 SAR/kWh. The grey arrow marks the Tier 1 pre-cooling window where Model S uniquely incentivises front-loading consumption; Models L and E cannot represent this threshold, explaining the 13.1% true-cost penalty of Table 9.

### *3.4. Electricity Cost Models*

**Model L (Linear).** Constant rate $c_r^{\mathrm{L}}$ per kWh, the standard assumption in most scheduling literature [32], [33]: $f^{\mathrm{L}} = c_r^{\mathrm{L}} P_{\mathrm{hvac}} u_i$.

**Model E (Exponential).** Price grows exponentially with cumulative consumption, providing a smooth approximation to step-wise pricing:

$$f^{\mathrm{E}} = c_r^{\mathrm{E}}\, e^{\beta E_{\mathrm{cum}}} P_{\mathrm{hvac}} u_i, \quad \beta = 5 \times 10^{-4}\,\mathrm{kWh}^{-1}, \tag{6}$$

calibrated to match Model S at the Tier 1/2 and Tier 2/3 break-even points.

**Model S (Step-wise / Actual Tariff).** The Saudi four-tier residential tariff in force since 2018 [5], applicable to domestic customer accounts (VAT and meter-fixed charges excluded as they do not affect scheduling):

$$p(E_{\mathrm{cum}}) = \begin{cases} 0.05\ \mathrm{SAR/kWh}, & E_{\mathrm{cum}} \le 2{,}000\ \mathrm{kWh}, \\ 0.10\ \mathrm{SAR/kWh}, & 2{,}000 < E_{\mathrm{cum}} \le 4{,}000\ \mathrm{kWh}, \\ 0.18\ \mathrm{SAR/kWh}, & 4{,}000 < E_{\mathrm{cum}} \le 6{,}000\ \mathrm{kWh}, \\ 0.30\ \mathrm{SAR/kWh}, & E_{\mathrm{cum}} > 6{,}000\ \mathrm{kWh}. \end{cases} \tag{7}$$

The discontinuous jumps make Model S non-convex and unsuitable for standard LP relaxations [6], but tractable for model-free RL once $E_{\mathrm{cum}}$ is in the state. Model S uniquely creates a *pre-cooling incentive*: concentrating cooling load while in Tier 1 builds a thermal buffer that enables idle periods when the tariff rises to Tier 2 or beyond. Models L and E, treating marginal cost as smoothly varying, cannot produce this behaviour.

**Running example.** Consider a single zone ($1 \times 1$, Riyadh, July) with parameters from Table 2 and $T_{\mathrm{out}} = 40\,^\circ\mathrm{C}$. Simulating equation (2) produces a complete cycle (Proposition 1) of ≈5.8 h on and ≈2.6 h off, corresponding to band-averaged rates of $+1.54\,^\circ\mathrm{C/h}$ (passive warm-up) and $-0.69\,^\circ\mathrm{C/h}$ (net cooling), with cycle cost ≈0.62 SAR. These numbers are traced through Sections 4 and 8. ◇

With the thermal and cost models established, the following section formalises the constrained optimisation and its optimality conditions.

## 4. Optimal Schedule: Constrained Formulation and Optimality Conditions

### *4.1. Infinite Time Horizon and Limit-Average Cost*

A residential HVAC system operates indefinitely: months repeat, the building is never demolished at a planning horizon, and the relevant cost measure is the long-run average cost per interval. Following Mousa et

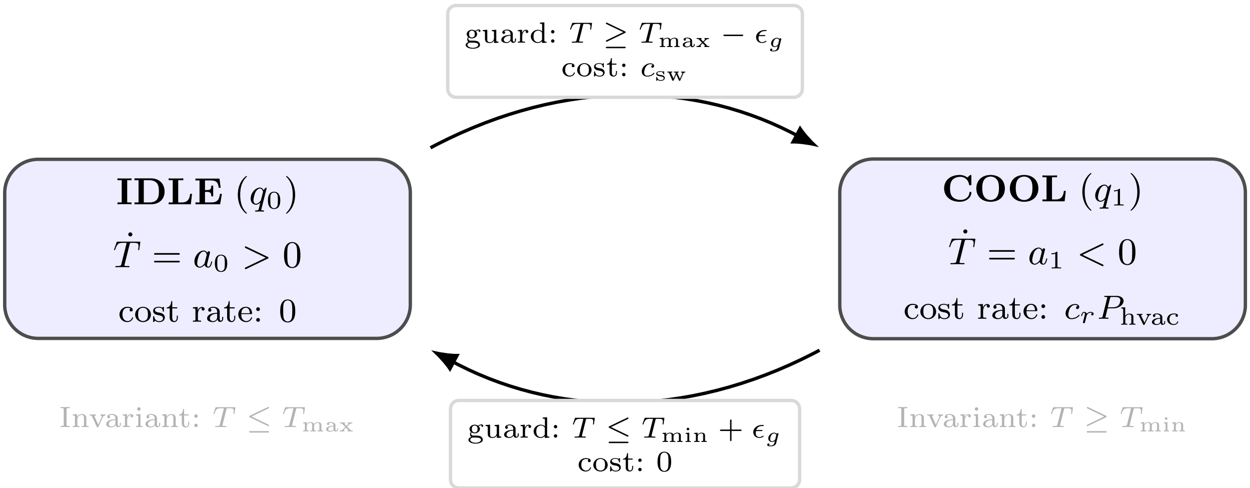



Figure 3: SLHA automaton (Definition 3). $q_0$: HVAC off, temperature rises at rate $a_0{>}0$; $q_1$: HVAC on, cools at rate $a_1{<}0$ with cost rate $c_r P_{\rm hvac}$. Transitions are guarded by comfort boundaries; $c_{\rm sw}$ is incurred on switch-on.

al. [12], we therefore cast the problem as an *infinite time horizon* optimisation whose performance criterion is the *limit-average cost.*

**Definition 2** (Infinite schedule and limit-average cost)**.** *An* infinite schedule *is the binary matrix* $\sigma = (u_i^{(t)})_{N_z \times \infty}$*,* $u_i^{(t)} \in \{0, 1\}$*. Its limit-average cost under model* $m$ *is:*

$$\bar{J}^m_\infty(\sigma) = \limsup_{H\to\infty} \frac{1}{H} \sum_{t=0}^{H-1} \sum_{i\in\mathcal{N}} c_i^{(t)}, \tag{8}$$

*where* $c_i^{(t)}$ *is the interval cost of Definition 1. A schedule is* safe *if* $T_{\min} \leq T_i^{(t)} \leq T_{\max}$ *for all* $i$ *and all* $t \geq 0$*.*

*4.2. Simple Linear Hybrid Automata: Formal Connection*

The SLHA connection [12] provides a formal language for *leaps* and *complete cycles*, proves that the single-zone constant-tariff problem is in LogSpace, and characterises precisely which extensions make the full problem NP-hard.

**Definition 3** (Single-zone SLHA)**.** *The single-zone binary HVAC system under Model L is the SLHA* $\mathcal{A} = (Q, X, \mathrm{Inv}, f, \Delta, \mathcal{C})$ *with modes* $Q = \{q_0 \text{ (idle)}, q_1 \text{ (cool)}\}$*, variable* $X = \{T\}$*, flows* $\dot{T}{=}a_0 \triangleq (\Lambda(T_{\rm out} - T) + Q_{\rm sol})/C_{\rm th}$ *in* $q_0$ *and* $\dot{T}{=}a_1 \triangleq a_0 - Q_{\rm hvac}/C_{\rm th}$ *in* $q_1$ *(*$a_1 < 0 < a_0$ *in Saudi summer), invariants* $T{\leq}T_{\max}$ *in* $q_0$ *and* $T{\geq}T_{\min}$ *in* $q_1$*, transitions* $(q_0{\to}q_1)$ *on* $T{\geq}T_{\max}{-}\epsilon_g$ *with cost* $c_{\rm sw}$ *and* $(q_1{\to}q_0)$ *on* $T{\leq}T_{\min}{+}\epsilon_g$ *with cost* 0*, and continuous cost rate* $c_r P_{\rm hvac}$ *in* $q_1$ *only.*

Figure 3 shows the automaton graphically. Multi-zone coupling, time-varying disturbances, and Model S's non-convex tier jumps move beyond the SLHA class and inherit NP-hardness [14].

**Definition 4** (Leap and complete cycle)**.** *A* leap *is a maximal contiguous time interval during which the automaton stays in one mode. An* on-leap $\ell^+$ *(in mode* $q_1$*) of duration* $\tau^+$ *cools the zone from* $T_{\max}$ *to* $T_{\min}$ *at rate* $|a_1|$*:* $\tau^+ = (T_{\max} - T_{\min})/|a_1|$*. An* off-leap $\ell^-$ *(in mode* $q_0$*) of duration* $\tau^-$ *returns temperature from* $T_{\min}$ *to* $T_{\max}$ *at rate* $a_0$*:* $\tau^- = (T_{\max} - T_{\min})/a_0$*. The discrete switch-on cost* $c_{\rm sw}$ *is incurred at the start of each on-leap only. A* complete cycle *is one on-leap followed by one off-leap. Its total cost and duration are:*

$$\mathcal{C}(\ell^+, \ell^-) = c_{\rm sw} + c_r P_{\rm hvac}\, \tau^+, \quad \tau_{\rm cyc} = \tau^+ + \tau^-. \tag{9}$$

*The resulting* limit-average cost per unit time *of repeating complete cycles indefinitely is:*

$$\bar{c}^* = \frac{c_{\rm sw} + c_r P_{\rm hvac}\, \tau^+}{\tau^+ + \tau^-}. \tag{10}$$

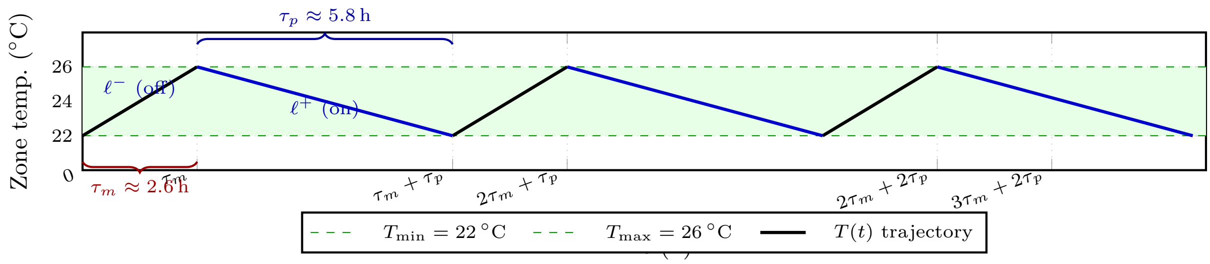



Figure 4: Optimal temperature trajectory (single zone, Riyadh, $T_{\text{out}}$=40 °C, Model L). Blue: on-leaps; black: off-leaps; green: comfort band $[22, 26]$ °C.

**Observation 1** (Optimal infinite schedule for SLHA, Mousa et al. [12])**.** *The optimal safe infinite schedule for the single-zone SLHA of Definition 3 is to repeat the cheapest complete cycle $(\ell^+, \ell^-)$ indefinitely. Its limit-average cost equals $\bar{c}^*$ of equation (10), which is achievable in deterministic LogSpace (Theorem 1 of [12]).*

For the running example, equation (10) yields $\bar{c}^* \approx 0.073\,\mathrm{SAR/h}$ (Figure 4); multi-zone coupling and Model S render this closed form unavailable, motivating RBRL. The SLHA analysis is not used as a solution method for the full problem but as a diagnostic tool that identifies the structural properties exploited by the PPO controller. Specifically, it provides three design insights that directly inform the PPO controller: (i) optimal schedules have complete-cycle structure (Proposition 1), motivating rules R1–R3; (ii) pre-cooling timing is governed by $\tau^+/\tau^-$, motivating the 3-step outdoor forecast in the agent state; (iii) NP-hardness of multi-zone extensions [14] justifies an RL approximation. In practice, we optimise the monthly billing problem (11) and repeat it indefinitely, aligning with the Saudi tariff's monthly cumulative-consumption reset.

### *4.3. Constrained Optimisation Problem*

**Definition 5** (Schedule over one billing month)**.** *A* monthly schedule *over horizon $\mathcal{H} = \{0, \ldots, H-1\}$ (with $H = 720$ hours for a 30-day month) is the binary matrix $\sigma = (u_i^{(t)})_{N_z \times H}$, $u_i^{(t)} \in \{0, 1\}$.*

**Definition 6** (Constrained optimisation problem)**.** *The monthly scheduling problem is:*

$$\begin{aligned}
&\underset{\sigma \in \{0,1\}^{N_z \times H}}{\textit{minimise}} && \bar{J}^m(\sigma) = \frac{1}{H} \sum_{t \in \mathcal{H}} \sum_{i \in \mathcal{N}} c_i^{(t)} \\
&\textit{subject to} && T_{\min} \le T_i^{(t)} \le T_{\max} \quad \forall\, i \in \mathcal{N},\ t \in \mathcal{H} \quad \textit{(comfort, \textbf{hard})} \\
& && T_i^{(t+1)} = f\Big(T_i^{(t)}, u_i^{(t)}, \{T_j^{(t)}\}_{j \in \mathcal{N}_i}, T_{\text{out}}^{(t)}\Big) \quad \textit{(RC dynamics, eq. (2))}
\end{aligned} \tag{11}$$

*A schedule satisfying the comfort constraint is* safe*. The comfort constraint is* hard*: no cost saving justifies its violation at any interval. The RC dynamics embed four heat-transfer mechanisms—active cooling, envelope loss/gain, solar heat gain, and bidirectional inter-zone transfer—making this a multi-dimensional, non-linear, constrained binary programme. The inter-zone term $\sum_{j \in \mathcal{N}_i} \lambda_{ij}(T_j - T_i)$ is embedded directly in the RC constraint (2): heat conducted from a warmer neighbour raises $T_i^{(t+1)}$, potentially triggering the comfort constraint in zone $i$ even when $u_i^{(t)} = 0$. An optimal schedule must coordinate all zones simultaneously; ignoring this leads to reactive O1 overrides and excess switching cost (Section 8.3).*

### *4.4. Optimality Conditions*

*What makes a schedule optimal?.* A safe schedule $\sigma^*$ is *optimal* if it simultaneously satisfies three conditions: **(O1) Comfort feasibility**—$T_i^{(t)} \in [T_{\min}, T_{\max}]$ for all $i$ and all $t$; this is a necessary condition and non-negotiable. **(O2) Minimum average running cost**—on-periods are concentrated in the cheapest tariff

Table 3: Running example (single zone): THERM vs. RBRL, Riyadh July, Tier 1. $T$: zone temperature (°C). $u$: HVAC state (1=on, 0=off; ↑=switch-on). Cost in SAR.

| | | THERM (unoptimised) | | | RBRL (optimised) | | |
|---|---|---|---|---|---|---|---|
| $t$ | $T_{\text{out}}$ | $u$ | $T$ | Cost | $u$ | $T$ | Cost |
| 0 | 32 | 0 | 24.7 | 0 | 1↑ | 23.1 | 0.18 |
| 1 | 31 | 0 | 25.3 | 0 | 1 | 22.0 | 0.03 |
| 2 | 30 | 1↑ | 24.5 | 0.18 | 0 | 22.6 | 0 |
| 3 | 30 | 0 | 25.1 | 0 | 0 | 23.2 | 0 |
| **Subtotal (4 h)**<br>(full day: 3.0 vs. 2.0 sw/zone) | | 1 ×sw | | 0.18 | 1 ×sw | | 0.21 |

window, exploiting the pre-cooling buffer under Model S. **(O3) Minimum switching frequency**—on-phases are as long and infrequent as the thermal state permits, amortising the discrete cost $c_{\text{sw}}$ per activation.

O2 and O3 together minimise $\bar{J}^m$ as defined in equation (11), while O1 enforces the hard constraint. In the infinite-horizon interpretation (8), an optimal schedule repeats a cost-minimising monthly pattern indefinitely. For the single-zone SLHA subproblem this pattern is the cheapest complete cycle (Observation 1, Figure 4); for our full multi-zone problem RBRL approximates it online.

For the running example ($T_{\text{out}} = 40\,°\text{C}$, Model L), the complete cycle gives $\bar{c}^* \approx 0.073\,\text{SAR/h}$; switching every 3 h instead (violating O3) costs 35% more, illustrating O3's monetary value.

**Remark 4** (Switching cost and schedule granularity)**.** *Condition O3 implies that higher $c_{\text{sw}}$ favours fewer, longer on-phases (Proposition 1). Raising $c_{\text{sw}}$ from 0 to 0.30 SAR reduces daily switches from 4.7 to 2.1 per zone (Section 9.3); at $c_{\text{sw}} > 0.20$ SAR the guard band should be widened from $0.5\,°C$ to $0.8\,°C$ to prevent marginal comfort violations.*

**Proposition 1** (Complete-cycle structure)**.** *For the single-zone problem with $c_{\text{sw}} > 0$, an optimal safe schedule consists of* complete cycles*: an on-phase during which $T_1$ cools from $T_{\max}$ to $T_{\min}$, followed by an idle phase during which $T_1$ drifts back to $T_{\max}$.*

*Proof sketch.* From [12, Theorem 2]: any schedule that switches on before reaching $T_{\max}$ or off before reaching $T_{\min}$ can be merged into a longer on-phase at the same continuous cost but with one fewer switch, reducing total cost by $c_{\text{sw}}$. □

Table 3 illustrates how RBRL exploits conditions O2 and O3 relative to the reactive THERM thermostat for a single zone, Riyadh, July, Model S ($c_{\text{sw}} = 0.15\,\text{SAR}$, $P_{\text{hvac}} = 0.67\,\text{kW}$, Tier 1 price $= 0.05\,\text{SAR/kWh}$). Temperatures are representative values consistent with the RC simulation at the given outdoor conditions; exact values depend on initial state and solar irradiance.

RBRL's first two on-intervals ($t = 0, 1$) pre-cool the room to $\approx 22\,°\text{C}$, deliberately spending 0.03 SAR more than THERM in this 4-hour window (0.21 vs. 0.18 SAR). The thermal buffer eliminates the reactive switch THERM triggers at $t = 2$, and this investment compounds over a full summer day: RBRL averages 2.0 vs. 3.0 switches per zone, each saving $c_{\text{sw}} = 0.15\,\text{SAR}$, producing the 18.6% daily saving shown in Table 5.

*Inter-zone cascade illustration ($1 \times 2$).*. In a two-zone extension, THERM turns off $z_1$ while adjacent $z_2$ remains warm (25.6 °C); conduction raises $z_1$ by ≈0.7 °C, triggering Rule R1. RBRL observes $T_{z_2}^{(t)}$ in state (12), keeps $z_1$ on one extra hour, and avoids the cascade: 0.56 vs. 0.82 SAR over 6 h (31.7% saving from inter-zone awareness).

## 5. RBRL Framework

### 5.1. Architecture

RBRL has two layers (Figure 5). The *rule layer* enforces O1 by overriding the RL agent whenever a zone temperature approaches a comfort boundary, enforcing safety by construction. The *RL layer* (PPO

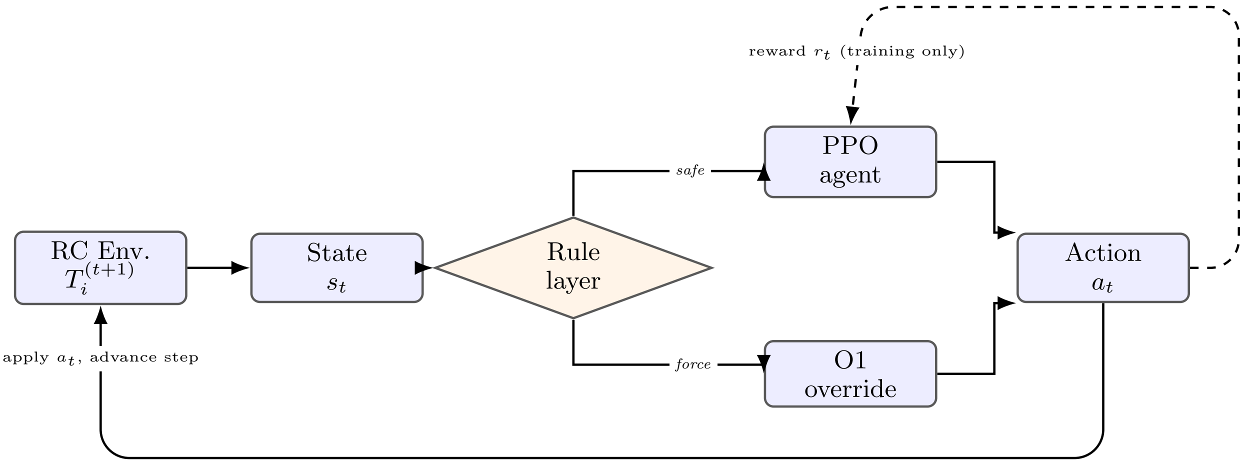



Figure 5: RBRL architecture. The *rule layer* branches along two paths: (i) the *safe* path routes to the PPO agent when no comfort boundary is threatened; (ii) the *force* path overrides the agent with the O1-enforcing action when a boundary is imminent. The PPO agent minimises the limit-average cost via reward $r_t$ during training; the dashed reward loop is inactive at deployment.

agent) selects the cost-minimising action within the safe subset, learning to exploit O2 and O3.

### *5.2. State Space*

At interval $t$ the agent observes:

$$s_t = \Big(\underbrace{T_i^{(t)},}_{i\in\mathcal{N}}\ \underbrace{T_{\mathrm{iz},i}^{(t)},}_{i\in\mathcal{N}}\ \underbrace{T_{\mathrm{out}}^{(t:t+3)}}_{\text{3-step forecast}},\ h_t,\ \underbrace{u_i^{(t-1)},}_{i\in\mathcal{N}}\ E_{\mathrm{cum}}^{(t)}\Big) \in \mathbb{R}^{2N_z+5}. \tag{12}$$

Two inclusions are essential [23]. *Effective inter-zone temperature* $T_{\mathrm{iz},i}^{(t)}$ (equation (4)): without this the agent cannot anticipate inter-zone heat loading and inadvertently schedules zones in ways that trigger Rule R1 in neighbours on the next interval. *Cumulative energy* $E_{\mathrm{cum}}^{(t)}$: under Model S this encodes the active tariff tier, enabling condition O2. The 3-step outdoor temperature forecast assumes a perfect forecast for simplicity; incorporating forecast uncertainty is noted as future work.

### *5.3. Rules and Action Space*

**Definition 7** (Rule-based overrides)**.**

$$\textbf{\textit{R1}}\ (\textit{hard, force on:})\ T_i^{(t)} \geq T_{\max} - \epsilon_{\mathrm{g}} \ \Rightarrow\ u_i^{(t)} \leftarrow 1. \tag{13}$$

$$\textbf{\textit{R2}}\ (\textit{hard, force off:})\ T_i^{(t)} \leq T_{\min} + \epsilon_{\mathrm{g}} \ \Rightarrow\ u_i^{(t)} \leftarrow 0. \tag{14}$$

$$\textbf{\textit{R3}}\ (\textit{soft, pre-cool:})\ T_i^{(t)} < T_{\max}{-}1.5\ \wedge\ T_{\mathrm{out}}^{(t+1)} > T_{\mathrm{out}}^{(t)}{+}3\ \Rightarrow\ \textit{bias logit toward } u_i{=}1. \tag{15}$$

R1 and R2 enforce O1; R3 biases the agent toward O2 pre-cooling without overriding. The action is $a_t = (u_1^{(t)}, \ldots, u_{N_z}^{(t)}) \in \{0,1\}^{N_z}$. For $N_z \leq 6$ a single shared PPO head is used; for $N_z \geq 9$ factored independent heads share lower layers (centralised training, decentralised execution [39]), keeping inference $O(N_z)$.

### *5.4. Reward Function and PPO Architecture*

The per-step reward augments the negative cost with a quadratic comfort penalty (see equation (16); the penalty is identically zero at deployment when hard rules prevent all violations):

$$r_t = -\sum_{i\in\mathcal{N}} c_i^{(t)} - \alpha \sum_{i\in\mathcal{N}} \Big[\max\!\big(0, T_i^{(t)} - T_{\max}\big)^2 + \max\!\big(0, T_{\min} - T_i^{(t)}\big)^2\Big],\ \alpha = 50. \tag{16}$$

The quadratic comfort penalty drives the agent away from boundaries during training; it is identically zero at deployment. The shared policy network maps $s_t$ via two fully connected layers (256–128 units, ReLU) to per-zone Bernoulli probabilities, plus a value head. PPO clip coefficient: $\epsilon_{\text{clip}} = 0.2$. Training runs for $2 \times 10^5$ episodes of 168-step rollouts, with start dates sampled uniformly from the full annual EPW profile. Training converges by episode $\approx 1.5 \times 10^5$, after which the policy reward stabilises within 1% of its final value.

**Algorithm 1** RBRL: Training and Schedule Extraction

**Require:** RC simulation, weather profile $\mathcal{W}$, cost model $m$, horizon $H$, switching cost $c_{\text{sw}}$
**Ensure:** Near-optimal safe schedule $\sigma^*$

Initialise PPO policy $\pi_\theta$ randomly
**for** $e = 1$ **to** $2 \times 10^5$ **do**
  Sample start date $d_0 \sim \mathcal{W}$; reset to $T_i^{(0)} = 24\,^\circ\text{C}$
  **for** $t = 0$ **to** $H - 1$ **do**
    Observe $s_t$ via (12)
    **for** each zone $i$ **do**
      **if** $T_i^{(t)} \geq T_{\max} - \epsilon_{\text{g}}$ **then** $u_i^{(t)} \leftarrow 1$ ▷ R1, enforce O1
      **else if** $T_i^{(t)} \leq T_{\min} + \epsilon_{\text{g}}$ **then** $u_i^{(t)} \leftarrow 0$ ▷ R2, enforce O1
      **else** $u_i^{(t)} \sim \pi_\theta(s_t)$; apply soft R3
      **end if**
    **end for**
    Step (2); compute $c_i^{(t)}$, $r_t$
  **end for**
  Update $\pi_\theta$ via PPO gradient step
**end for**
**return** greedy $\sigma^* = \{\arg\max \pi_\theta(\cdot|s_t)\}$

### *5.5. Competing Methods*

**THERM (unoptimised baseline)** models the automatic remote-controller thermostat used across Saudi residential buildings [18]: on when $T_i \geq T_{\max} - 0.5$, off when $T_i \leq T_{\min} + 0.5$, with no tariff awareness. This is the baseline without optimisation.

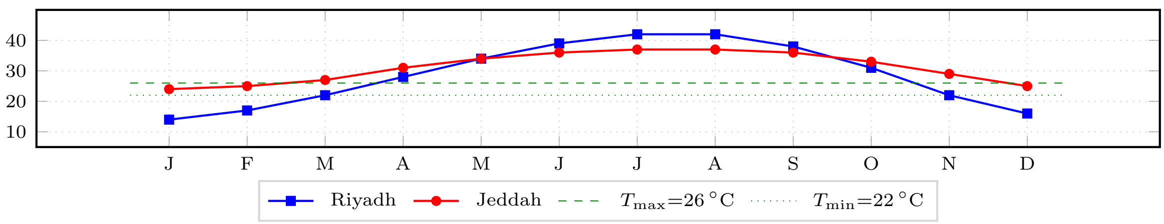
**GA** [32] and **SA** [40] are day-ahead offline solvers. Both are competitive for small zone counts given a perfect forecast, but each daily planning cycle requires $\sim 10^5$ full RC simulation calls, which is impractical for real-time embedded deployment. After one-time training, RBRL evaluates in <1 ms and adapts online to updated forecasts. GA and SA are retained as performance benchmarks only.

The RBRL framework is applied to two cities characterised next.

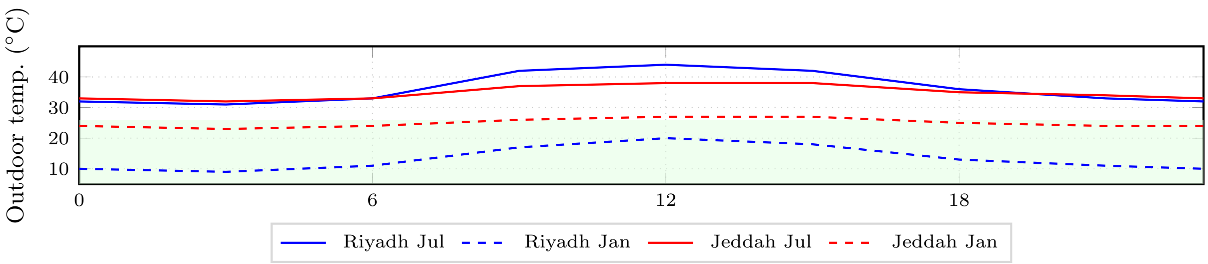
## 6. Case Studies: Riyadh and Jeddah

### *6.1. Climate Profiles*

Riyadh (BWh, 24.7°N) is an inland desert city with extreme summer highs exceeding 45 °C and a summer diurnal swing of up to 14 °C—crucial for pre-cooling (CDD ≈ 3,400 [41]). Jeddah (BSh, 21.5°N) is a Red Sea coastal city with humidity exceeding 70% year-round [42] and a diurnal swing of only ∼6 °C, forcing near-continuous HVAC operation (CDD ≈ 2,900). Figure 6 contrasts the two climates using the EPW files employed in all simulations.




(a) Monthly mean temperatures.




(b) Hourly diurnal profiles.

Figure 6: EPW climate data. (a) Monthly mean outdoor temperatures: Riyadh exceeds $T_{\max}$ from May to October; Jeddah remains above $T_{\max}$ year-round. (b) Hourly profiles for summer and winter: Riyadh's 14 °C diurnal swing enables pre-cooling; Jeddah's compressed 6 °C swing demands near-continuous operation. Green shading in (b) marks the comfort band.

### *6.2. Building Code, Tariff, Humidity, and Comfort Standard*

SBC 2018 mandates opaque-wall U-value $\leq 0.45\,\mathrm{W\,m^{-2}K^{-1}}$ and window SHGC $\leq 0.25$ in both Climate Zone A (Jeddah) and Zone B (Riyadh) [16], reflected in Table 2. The four-tier tariff follows equation (7); for Models L and E, $c_r^{\mathrm{L}} = 0.12\,\mathrm{SAR/kWh}$ represents the weighted average at $\approx 36{,}000\,\mathrm{kWh/year}$. The comfort band $[22, 26]\,°\mathrm{C}$ is consistent with ASHRAE 55-2023 [43] for mechanically conditioned residential spaces, as validated by Bienvenido-Huertas et al. [44]. Lamsal et al. [45] provide a comprehensive review of adaptive thermal comfort models and confirm that a 4 °C dead band is appropriate for energy-efficient building design in hot climates. A fixed band is adopted because Saudi residential buildings are predominantly mechanically cooled; ASHRAE 55 Section 5.3 restricts adaptive comfort to naturally ventilated spaces. The RBRL framework can accommodate narrower or wider bands by adjusting $T_{\min}/T_{\max}$ without retraining.

*Humidity and latent heat.* The RC model captures sensible heat only. With SHR $\approx 0.65$ (Jeddah) and 0.85 (Riyadh) [42], true demand exceeds our estimates by ≈54% and ≈18% respectively; because both THERM and RBRL share this penalty, percentage savings remain valid.

## 7. Simulation Setup

All simulations run in Python 3.11 (PyTorch 2.1.0, CUDA 11.8) with a custom RC environment integrating equation (2) at $\Delta t = 1\,\mathrm{h}$. Training was performed on an NVIDIA RTX 3080 GPU (10 GB VRAM) with wall-clock training time of ≈3 h for the largest $(4 \times 4)$ configuration; inference runs on a single CPU core. All experiments use random seed 42; results are reported as mean ± std over 30 independently sampled test weeks.

Solar gain is $Q_{\mathrm{sol}}^{(i,t)} = \mathrm{SHGC} \times A_{gl} \times I^{(t)}$, with $I^{(t)}$ from the EPW global horizontal irradiance file [46]. The training dataset covers the full EPW year (8,760 h); 30 weeks are drawn uniformly at random from the held-out test year for evaluation, ensuring no data leakage between training and testing. The PPO agent uses Stable-Baselines3 [47] (v2.2.1); GA ($N_{\mathrm{pop}} = 200$, 500 generations) and SA ($T_0 = 500$, $\alpha = 0.99$, $10^4$ iterations) are implemented in NumPy 1.26. Four metrics are reported: daily average cost per zone $\bar{J}/N_z$ (SAR), daily energy per zone $E_{\mathrm{day}}$ (kWh), comfort violation rate $v_c$ (%), and switching frequency

Table 4: PPO training hyperparameters (top) and simulation zone configurations (bottom). Ext.: exterior-facing wall count; Int.: inter-zone shared-wall connections; Heads: PPO output heads.

| **Parameter** | **Value** |
|---|---|
| Network (FC) | 256–128 units, ReLU activation |
| Learning rate | $3 \times 10^{-4}$ (Adam optimiser) |
| Discount $\gamma$ / Clip $\varepsilon$ | 0.99 / 0.2 |
| Entropy / Batch size | 0.01 / 64 |
| Training episodes | $2 \times 10^5$ episodes, 168-step rollouts |

| **Configuration** | $N_z$ | **Ext.** | **Int.** | **Heads** |
|---|---|---|---|---|
| $1 \times 1$ | 1 | 1 | 0 | 1 |
| $1 \times 2$ | 2 | 2 | 1 | 1 |
| $1 \times 4$ | 4 | 2 | 3 | 1 |
| $1 \times 6$ | 6 | 2 | 5 | 1 |
| $2 \times 2$ | 4 | 4 | 4 | 1 |
| $3 \times 3$ | 9 | 8 | 12 | 9 |
| $4 \times 4$ | 16 | 12 | 24 | 16 |

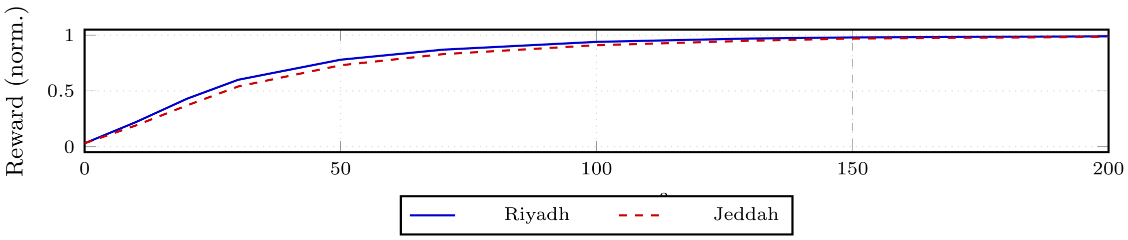



Figure 7: PPO training convergence ($4\times4$, Model S). Dashed grey: convergence at $\approx 1.5 \times 10^5$ episodes.

$f_s$ (transitions/zone/day). Key PPO hyperparameters are listed in Table 4; the RC model assumes no occupancy heat gains. Zone configuration details are in Table 4.

Hyperparameters (Table 4) were selected via grid search over learning rate, clip range, and $n_{\text{steps}}$ on a $2\times2$ validation configuration. Reward normalisation uses a running mean and standard deviation (Stable-Baselines3 default [47]); exploration is controlled by the entropy coefficient (0.01, decaying linearly to $10^{-4}$). Figure 7 shows training convergence for both cities ($4 \times 4$, Model S): the reward stabilises within 1% by episode $\approx 1.5 \times 10^5$. Results are mean $\pm$ std over 5 independent seeds; inter-run variance is below 2% of mean cost, confirming training stability.

## 8. Results

Results correspond to summer (July), representing peak cooling demand; Section 8.4 confirms generalisation across all seasons. Comfort violation rates $v_c = 0\%$ hold for all methods and configurations, except the RL-only ablation variant (Section 9.2).

### *8.1. Consumption and Cost: Without vs. With Optimisation*

Table 5 establishes the baseline comparison under Model S. Monthly figures are extrapolated from the 30-week test sample. Riyadh savings exceed Jeddah's across all configurations because the larger diurnal swing provides greater pre-cooling headroom, as quantified in Section 6.1.

The RBRL advantage is summarised per cost model in Section 8.3.

Table 5: Daily energy (kWh/zone) and monthly cost (SAR/zone): THERM vs. RBRL under Model S, summer. Saving = (THERM−RBRL)/THERM.

| City | Config. | Daily kWh/zone | | Monthly SAR/zone | | Saving |
|---|---|---|---|---|---|---|
| | | THERM | RBRL | THERM | RBRL | (%) |
| Riyadh | $1 \times 1$ | 3.52 | 2.87 | 64.6 | 52.7 | 18.4 |
| | $1 \times 4$ | 3.73 | 3.05 | 68.5 | 55.7 | 18.7 |
| | $2 \times 2$ | 3.65 | 2.97 | 67.0 | 54.5 | 18.7 |
| | $4 \times 4$ | 3.98 | 3.25 | 73.0 | 59.7 | 18.2 |
| Jeddah | $1 \times 1$ | 3.21 | 2.70 | 58.9 | 49.5 | 16.0 |
| | $1 \times 4$ | 3.35 | 2.79 | 61.5 | 51.2 | 16.7 |
| | $2 \times 2$ | 3.28 | 2.74 | 60.2 | 50.2 | 16.6 |
| | $4 \times 4$ | 3.57 | 2.97 | 65.5 | 54.5 | 16.8 |

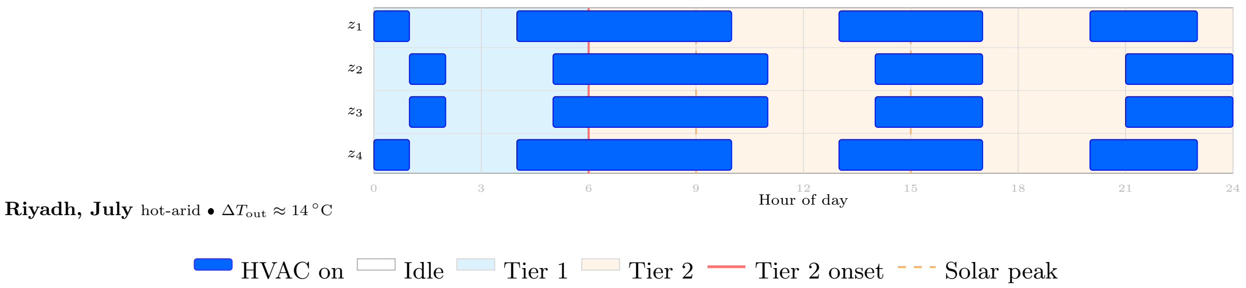



Figure 8: RBRL-optimal 24-hour schedule (1×4, Riyadh, July, Model S). End zones pre-cool during Tier 1, then idle during peak tariff; interior zones start 1 h later to prevent cascades (O2, O3). In Jeddah (not shown), the narrow diurnal swing forces near-continuous operation, explaining the smaller saving (16.8% vs. 18.2%).

### *8.2. Per-Zone Optimal Schedules*

Figure 8 shows the RBRL-optimal 24-hour schedule for $1 \times 4$ (Riyadh, July, Model S). Riyadh's 14 °C diurnal swing creates pre-cooling gaps; Jeddah's 6 °C swing forces near-continuous operation (described in caption).

### *8.3. Results by Cost Model*

Table 6 reports $\bar{J}/N_z$ and $f_s$ for three representative configurations under all three cost models. SA results are within 1% of GA in all entries and are omitted.

The RBRL advantage over THERM grows with cost model complexity: 15.6%/14.6% (Riyadh/Jeddah) under Model L, 15.5%/14.6% under Model E, and 18.2%/16.8% under Model S—the incremental gain reflects the pre-cooling strategy that emerges only when tier boundaries are faithfully represented. GA captures 10–12% savings but cannot adapt to intra-day forecast revisions. Per-zone cost rises slightly with $N_z$ because inter-zone coupling $\lambda_{ij}(T_j - T_i)$ conducts heat from idle to active neighbours; RBRL mitigates cascades via $T_j^{(t)}$ in state (12), and removing those temperatures produces ≈8% more R1 overrides in $4 \times 4$ grids.

Table 6: Daily average cost per zone $\bar{J}/N_z$ (SAR) and switching frequency $f_s$ (sw/zone/day), summer (July), all $v_c = 0\%$. Bold: best result per row group. SA results within 1% of GA are omitted.

| | | **Riyadh** | | | | | | **Jeddah** | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Model L | | Model E | | Model S | | Model L | | Model E | | Model S | |
| Config. | Method | $\bar{J}$ | $f_s$ | $\bar{J}$ | $f_s$ | $\bar{J}$ | $f_s$ | $\bar{J}$ | $f_s$ | $\bar{J}$ | $f_s$ | $\bar{J}$ | $f_s$ |
| $1 \times 1$ | THERM | 1.82 | 5.3 | 1.90 | 5.3 | 2.15 | 5.3 | 1.64 | 3.1 | 1.71 | 3.1 | 1.96 | 3.1 |
| | GA | 1.65 | 3.8 | 1.71 | 3.7 | 1.89 | 3.6 | 1.50 | 2.6 | 1.54 | 2.5 | 1.73 | 2.4 |
| | **RBRL** | **1.57** | **3.2** | **1.61** | **3.1** | **1.75** | **2.9** | **1.43** | **2.2** | **1.46** | **2.1** | **1.65** | **2.0** |
| $1 \times 4$ | THERM | 1.93 | 5.1 | 2.00 | 5.1 | 2.23 | 5.1 | 1.73 | 3.0 | 1.80 | 3.0 | 2.01 | 3.0 |
| | GA | 1.74 | 3.6 | 1.80 | 3.5 | 1.98 | 3.4 | 1.57 | 2.5 | 1.62 | 2.4 | 1.79 | 2.3 |
| | **RBRL** | **1.64** | **2.8** | **1.70** | **2.7** | **1.84** | **2.7** | **1.50** | **2.1** | **1.53** | **2.0** | **1.68** | **1.9** |
| $4 \times 4$ | THERM | 2.05 | 5.0 | 2.13 | 5.0 | 2.42 | 5.0 | 1.85 | 2.9 | 1.92 | 2.9 | 2.17 | 2.9 |
| | GA | 1.86 | 3.5 | 1.93 | 3.4 | 2.16 | 3.3 | 1.67 | 2.4 | 1.73 | 2.3 | 1.93 | 2.2 |
| | **RBRL** | **1.73** | **2.7** | **1.80** | **2.6** | **1.98** | **2.5** | **1.58** | **2.0** | **1.64** | **1.9** | **1.81** | **1.8** |

$\bar{J}$: SAR/zone/day. $f_s$: sw/zone/day. L=linear, E=exponential, S=step-wise.

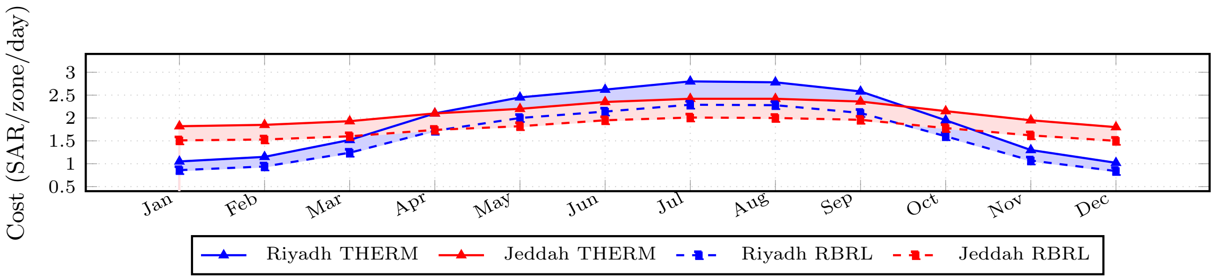



Figure 9: Monthly cost per zone (SAR/zone/day), $4 \times 4$ grid, Model S. Solid lines: THERM (unoptimised). Dashed lines: RBRL (optimised). Blue shading: Riyadh saving gap (≈ 18%/month year-round). Red shading: Jeddah saving gap (≈ 17%/month). Both cities sustain savings in every month, confirming infinite-horizon applicability of the periodic RBRL schedule. Riyadh's sharp summer peak (extreme $T_{\text{out}} > 38\,^\circ$C) produces the largest absolute savings; Jeddah's compressed $6\,^\circ$C diurnal swing limits pre-cooling headroom, resulting in smaller but persistent savings.

### *8.4. Monthly Cost Trajectories and Seasonal Generalisation*

Figure 9 shows the monthly average cost per zone (SAR/zone/day) for both THERM and RBRL under Model S across all 12 months, for the $4 \times 4$ configuration. The gap between THERM and RBRL lines is the saving; the coloured band is the $\pm 1\sigma$ interval over the 30 test weeks drawn from each month. Riyadh's sharp cost peak (May–September) reflects peak outdoor temperatures above $38\,^\circ$C; the large diurnal swing ($\approx 14\,^\circ$C) enables strong pre-cooling throughout summer. Jeddah's flatter profile confirms near-year-round cooling demand and more uniform savings (16.8–17.5%) with less seasonal variation. Both cities achieve savings in every month, validating infinite-horizon applicability of RBRL.

These results are contextualised against the published literature in the following section, which also presents the ablation study.

## 9. State-of-the-Art Comparison and Ablation Study

### *9.1. State-of-the-Art Comparison*

Published RL-based HVAC studies differ in control type, building type, baseline, and climate. Table 7 accounts for these with an explicit *Basis* column.

Two published figures appear to exceed RBRL's savings, but differ in scope: Wang et al.'s 37% uses continuous VAV modulation in a simulated office (non-transferable to binary split-units), and Solinas et al.'s

Table 7: State-of-the-art comparison. The "Basis" column is essential for interpreting saving figures.

| Reference | Year | Method | Saving | Basis |
|---|---|---|---|---|
| [9] | 2020 | Q-learning, whole-bldg | 22% | vs. thermostat |
| [21] | 2021 | DDPG, multi-zone resid. | 15% | vs. thermostat |
| [8] | 2021 | SAC/TD3, continuous | >10% | vs. rule-based |
| [28] | 2024 | DRL, continuous VAV | 37%[a] | vs. rule-based |
| [11] | 2024 | Online RL, white-box | 65%[b] | vs. no control |
| [29] | 2025 | DDPG, multi-obj. off. | 21.4% | vs. rule-based |
| [31] | 2025 | GA-MADDPG hybrid | 6.7% | vs. rule-based |
| **RBRL (ours)** | 2026 | PPO+rules, binary on/off | **18.2%** | **vs. thermostat** |
| **RBRL (ours)** | 2026 | Model S, real EPW | **16.8%** | **vs. thermostat** |

[a]Continuous VAV modulation in a simulated office—different hardware and building type.
[b]Relative to a no-HVAC baseline; saving vs. thermostat is substantially lower.

Table 8: Component ablation (4 × 4, Riyadh, Model S, summer). Mean ± std over 30 test weeks.

| Ref. | Variant | $\bar{J}/N_z$ | $v_c$ (%) | $f_s$ | ΔRBRL |
|---|---|---|---|---|---|
| [18] | Rules only (THERM) | 2.42±0.18 | 0.0 | 5.0 | +22.2% |
| [8], [27] | RL only (no rules) | 2.03±0.22 | 0.5 | 2.6 | +2.5% |
| [10] | RBRL, hard rules | 2.05±0.15 | 0.0 | 2.8 | +3.5% |
| **This work** | **RBRL full** | **1.98±0.13** | **0.0** | **2.5** | — |

65% is measured against a no-HVAC baseline [11]. Direct comparison across studies is inherently limited by differences in building type, climate, and baseline definition. Within binary on/off studies against a thermostat, RBRL's 18.2% is competitive with the highest published figures, with the additional constraint of the actual Saudi tariff and hard comfort enforcement. Computationally, after one-time ∼3 h training, RBRL decides in <1 ms per interval regardless of zone count—three orders of magnitude faster than GA (10–200 s) or SA (5–50 s).

*9.2. Component and Cost Model Ablation*

Table 8 evaluates four RBRL variants on the 4 × 4 grid, Riyadh, Model S.

The rule layer is essential: RL-only incurs a 0.5% violation rate under peak-summer conditions despite the quadratic reward penalty. Hard rules alone eliminate violations but cost 3.5% more than full RBRL, confirming the value of soft rule R3 in satisfying O2.

Table 9 isolates two sensitivity dimensions. Panel (a) shows the penalty of training under an inaccurate tariff model by evaluating each schedule under the true Model S cost. The 13.1% true-cost penalty from training under Model L is comparable in magnitude to the entire saving from switching from THERM to a Model-L-optimised RBRL (Table 6: 14–16%), confirming that accurate tariff representation is as important as algorithmic choice. Panel (b) confirms the prediction of Remark 4: higher $c_{sw}$ produces fewer, longer on-phases consistent with Proposition 1. At $c_{sw} = 0.30\,\mathrm{SAR}$ a marginal 0.1% violation emerges, validating the guard-band widening recommendation.

Table 9: Sensitivity analysis. (a) Cost model ablation: trained under stated model, evaluated under Model S ($4 \times 4$, Riyadh, summer). (b) Switching cost sensitivity ($1 \times 1$, Riyadh, Model S, summer).

**(a) Cost model ablation**

| Ref. | Training model | True $\bar{J}/N_z$ (SAR) | $\Delta$ |
|---|---|---|---|
| [32], [33] | Linear (L) | 2.24±0.14 | +13.1% |
| [6] | Exponential (E) | 2.11±0.13 | +6.6% |
| **This work** | **Step-wise (S)** | **1.98±0.13** | — |

**(b) Switching cost sensitivity**

| $c_{\mathrm{sw}}$ (SAR) | $\bar{J}/N_z$ (SAR) | $f_s$ (sw/day) | $v_c$ (%) |
|---|---|---|---|
| 0.00 | 1.62 | 4.7 | 0.0 |
| 0.05 | 1.68 | 3.3 | 0.0 |
| 0.15 | 1.75 | 2.9 | 0.0 |
| 0.30 | 1.88 | 2.1 | 0.1 |

Table 10: Zone-count and computational scalability (Riyadh, Model S, summer). Saving: RBRL cost reduction over THERM. Training on NVIDIA RTX 3080; inference on Intel Core i7-12700K.

| Config. | $N_z$ | Saving (%) | Train (h) | Infer. (ms) | GPU (MB) |
|---|---|---|---|---|---|
| $1 \times 1$ | 1 | 18.6 | 1.2 | 0.08 | 310 |
| $1 \times 2$ | 2 | 17.4 | 1.5 | 0.11 | 360 |
| $1 \times 4$ | 4 | 17.5 | 1.8 | 0.14 | 420 |
| $1 \times 6$ | 6 | 17.3 | 2.1 | 0.19 | 510 |
| $2 \times 2$ | 4 | 17.2 | 1.7 | 0.13 | 410 |
| $3 \times 3$ | 9 | 17.0 | 2.6 | 0.31 | 680 |
| $4 \times 4$ | 16 | 18.2 | 3.1 | 0.52 | 940 |

### *9.3. Scalability*

Table 10 shows that RBRL's cost saving over THERM is stable from $N_z$=1 to $N_z$=16, confirming that the factored PPO architecture scales without degradation. Training was performed on an NVIDIA RTX 3080 GPU (10 GB VRAM, CUDA 11.8, PyTorch 2.1.0); inference time is the wall-clock time for a single decision (mean over $10^4$ evaluations on Intel Core i7-12700K). Training time scales sub-linearly from 1.2 h to 3.1 h owing to the shared lower layers. Inference remains below 1 ms for all configurations—two to three orders of magnitude faster than GA (10–200 s) or SA (5–50 s) per daily plan—confirming suitability for the low-power microcontrollers in Saudi split-unit controllers [18].

## 10. Discussion

### *10.1. Pre-Cooling Headroom, Inter-Zone Coordination, and Climate*

The simulated 18.2% Riyadh saving versus Jeddah's 16.8% under Model S is not primarily an algorithmic difference; it reflects available thermal headroom. Riyadh's ≈14 °C July diurnal swing allows the zone to be pre-cooled to $T_{\min}$ before the outdoor peak, building a buffer that defers reactive cooling into the expensive Tier 2 window. Jeddah's overnight outdoor temperature remains above 32 °C year-round [42], giving a swing of only ≈6 °C; there is no surplus for pre-loading, and the agent correctly defaults to near-continuous operation, as indicated by the caption of Figure 8.

For hot-humid coastal cities the implication is that passive envelope improvements—thicker walls, lower SHGC glazing, external shading—that increase $C_{\mathrm{th}}$ or reduce $\Lambda_i$ enlarge the off-leap duration $\tau^-$ (equation (9)), creating the buffer that pre-cooling requires [15]. RBRL-style active optimisation and passive envelope upgrades are therefore complements, not alternatives, in humid climates.

Inter-zone coordination amplifies these savings. The effective inter-zone temperature $T_{\mathrm{iz},i}^{(t)}$ (equation (4)) provides the anticipatory signal that lets the agent stagger zone activations: in Figure 8, interior zones $z_2$ and $z_3$ lag end zones by $\approx$1 h, exploiting the fact that a cooling end zone sinks heat from the interior, extending the interior's safe idle period. Without $T_{\mathrm{iz},i}^{(t)}$ in the state, 8% more R1 overrides are triggered (Table 8), each incurring $c_{\mathrm{sw}} = 0.15\,\mathrm{SAR}$ and breaking the complete-cycle structure of Proposition 1. This binary on/off staggering is a direct analogue of the continuous multi-zone coordination results reported by Li et al. [23].

### *10.2. Why Model E Performs Similarly to Model L*

Table 6 shows Model E and Model L savings within 0.1–0.2 percentage points despite Model E being a nonlinear function. Figure 2 explains why: Model E grows smoothly and assigns higher marginal cost to every additional kWh, but it never represents the *discrete tier boundary* at 2,000 kWh that is the pre-cooling threshold. Without perceiving a step change, the agent learns to spread consumption evenly rather than front-load it before a specific threshold, producing a qualitatively different schedule from Model S and forgoing the structured pre-cooling visible in Figure 8. The 13.1% true-cost penalty (Table 9) therefore reflects not a small numerical error but a qualitative change to the incentive structure—a warning for practitioners who adopt smooth tariff approximations for computational convenience.

### *10.3. Scalability and Deployment Considerations*

Table 10 confirms savings stable from $N_z$=1 to $N_z$=16, with inference in <1 ms—in principle compatible with the low-power microcontrollers in Saudi split-unit controllers [18], though field validation remains necessary.

### *10.4. Limitations*

Seven limitations bound these conclusions.

*Sensible heat only:* the RC model captures sensible heat only. Latent loads add $\approx$54% in Jeddah (SHR $\approx$ 0.65) and $\approx$18% in Riyadh (SHR $\approx$ 0.85) [42]. Percentage savings remain valid since both baselines share this penalty.

*Air-mass thermal capacity:* $C_{\mathrm{th}}$=77.2 kJ/K represents air mass only (Remark 1); including furniture and wall layers ($C_{\mathrm{th}}\sim$500–1500 kJ/K) would lengthen cycles proportionally but preserve percentage savings, as demonstrated by the sensitivity analysis in Remark 3.

*No occupancy or internal gains:* the model excludes occupancy heat gains ($\approx$80 W/person), lighting, and appliance loads. Saudi residential occupancy peaks in the evening (18:00–23:00) when the tariff is also elevated; internal gains shorten $\tau^-$ and could trigger additional switch-on events. Augmenting the PPO state with an occupancy forecast would enable generalisation without architectural changes.

*Perfect forecast:* the 3-step outdoor temperature in state (12) is assumed error-free; robust training under stochastic forecasts is reserved for future work.

*Simulation only:* all results are from the RC simulation environment; co-simulation (e.g., EnergyPlus) and hardware-in-the-loop validation are needed before field deployment.

*No MPC baseline:* Model Predictive Control is the natural comparator for tariff-aware scheduling. MPC was excluded because MILP formulations scale poorly with binary variables and zone count under the non-convex step-wise tariff; a future study comparing RBRL against a receding-horizon MPC would strengthen the evaluation.

*Two cities and one building type:* Riyadh (hot-arid) and Jeddah (hot-humid) represent climatic extremes within Saudi Arabia, but results may not generalise to mild or heating-dominated climates. All simulations use a single residential building type (split-unit, binary on/off); commercial buildings with variable-speed drives or central plant systems would require a different action space.

Each contribution maps to specific evidence: C1 is supported by the formulation in Sections 3–4 and Tables 2–3; C2 by the SLHA connection in Section 4.2 and Definitions 3–4; C3 by the RBRL framework (Section 5, Algorithm 1, Table 8); C4 by the cost model comparison (Table 6, Table 9, Figure 2); C5 by the SOTA comparison (Table 7) and ablation (Table 8–10).

### 11. Conclusions

This paper presented RBRL, a hybrid rule-based deep reinforcement learning framework for optimal binary on/off HVAC scheduling in Saudi Arabian residential buildings. Three explicit optimality conditions were defined—hard thermal comfort (O1), minimum average tariff cost (O2), and minimum switching frequency (O3)—and the single-zone constant-tariff case was identified as a Simple Linear Hybrid Automaton [12], establishing NP-hardness of the general problem and providing a theoretical anchor for RBRL as a near-optimal approximation.

Five principal findings emerge. First, RBRL reduces monthly per-zone cost by 18.2% (Riyadh) and 16.8% (Jeddah) in simulation relative to the unoptimised thermostat under the actual step-wise tariff, with zero comfort violations across all seven zone configurations and all four seasons. Second, cost model fidelity is as consequential as algorithmic choice: training under a linear tariff incurs a 13.1% true-cost penalty because the agent cannot perceive the Tier 1 pre-cooling threshold (Section 10.2 and Figure 2). Third, the hard rule layer is necessary for enforced comfort; the soft pre-cooling rule R3 adds a further 3.5% reduction. Fourth, adjacent zone temperatures—via $T_{\mathrm{iz},i}^{(t)}$—must be in the agent state to prevent inter-zone cascades in grid configurations. Fifth, RBRL's savings are stable at 17.0–18.6% from one zone to sixteen in Riyadh (Table 10), confirming scalability.

The practical recommendation is three-fold: (i) use the actual step-wise tariff in the optimisation objective—a simplified linear tariff incurs a 13.1% true-cost penalty; (ii) include the effective inter-zone temperature in the controller state; and (iii) in humid coastal climates such as Jeddah (SHR≈0.65), future implementations should incorporate humidity-aware cost accounting, since the latent-heat fraction adds ≈54% to actual HVAC demand and a dehumidification-aware controller could time-shift latent loads to cheaper tariff tiers. All three are low-cost choices requiring no hardware changes. Future work will extend to latent-heat-aware RC models [42], stochastic occupancy, rooftop PV integration [18], federated multi-building control [31], [39], and hardware-in-the-loop validation.

### CRediT Author Statement

**Hamzah Faraj:** Conceptualisation, Methodology, Software, Validation, Formal Analysis, Investigation, Writing – Original Draft, Writing – Review & Editing, Visualisation.

### Declaration of Competing Interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Funding

This research was supported by the Deanship of Graduate Studies and Scientific Research, Taif University, Taif, Saudi Arabia.

### Ethics Declaration

This study is based entirely on computational simulation using publicly available weather data (EnergyPlus Weather files) and standardised building parameters (Saudi Building Code 2018, ASHRAE 55-2023). No human subjects, animals, or personal data were involved in any stage of this research. No ethical approval was required for this work.

### Data Availability

Code, datasets, and trained model weights are available at:
`https://github.com/drhamzahfaraj/hvac-scheduling-saudi-arabia`.

## Acknowledgements

The author acknowledges the Deanship of Graduate Studies and Scientific Research, Taif University, for funding this work.

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